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18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$n = 200$

$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$

H_a: at least one proportion is different

$E = \frac{200}{3}$

$= 66.67$

$\chi^2 = \Sigma \frac{(O-E)^2}{E}$

$= \frac{(83-66.67)^2 + (61-66.67)^2 + (56-66.67)^2}{66.67}$

$= 6.19$

$df = k-1$

$= 3-1$

$= 2$

$\chi^2_{0.05,2} = 5.991$

$\because 6.19 > 5.991$

$\therefore$ Reject H_0, there is enough evidence ...

$\hat{p} = \frac{83}{200}$

$= 0.415$

$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$0.415 \pm 1.96\sqrt{\frac{0.415(1-0.415)}{200}} = 0.415 \pm 1.96(0.0348)$

$= (0.347, 0.483)$

$\therefore (0.347, 0.483)$

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$n = 200$

$H_0 : P_{Sun} = P_{MON} = \cdots = P_{SAT} = \frac{1}{7}$

H_a : At least one proportion is different

$E = \frac{200}{7}$

$= 28.57$

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(24 - 28.57)^2 + (36 - 28.57)^2 + (27 - 28.57)^2 + (26 - 28.57)^2 + (32 - 28.57)^2 + (26 - 28.57)^2 + (29 - 28.57)^2}{28.57}$$

$= 3.63$

$df = k - 1$

$= 7 - 1$

$= 6$

$\chi^2_{0.05, 6} = 12.592$

$\because 3.63 < 12.592$

$\therefore$ Fail to reject H_0. there is no enough evidence that ...

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	
Fatal	67	26	16	109
Not Fatal	128	63	46	237
	195	89	62	

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

Ho : Fatal accidents is independent on the size of automobiles.
Ha : Fatal accidents is dependent on the size of automobiles.

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

	Small	Medium	Large
Fatal	61.43	28.04	19.53
Not Fatal	133.57	60.96	42.47

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= \frac{(67-61.43)^2}{61.43} + \frac{(128-133.57)^2}{133.57} + \frac{(26-28.04)^2}{28.04} + \frac{(63-60.96)^2}{60.96} + \frac{(16-19.53)^2}{19.53} + \frac{(46-42.47)^2}{42.47}$$

$$= 1.89$$

$$df = (r-1)(c-1)$$
$$= (2-1)(3-1)$$
$$= 2$$

$$\chi^2_{0.05,2} = 5.991$$

$\because$ $1.89 < 5.991$
$\therefore$ Non-reject Ho

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

DATA SET DS1403

	Workdays at Home per Week			
Salary	Less Than One	At Least One, but Not All	All at Home	
Under \$50,000	38	16	14	68
\$50,000–\$74,999	54	26	12	92
\$75,000–\$99,999	35	22	9	66
Above \$100,000	33	29	12	74
	160	93	47	300

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusions from part a?

a) Ho: The salary is independent on the number of workdays spent at home.
Ha: The salary is dependent on the number of workdays spent at home.

Salary	Less than One	At least one, but NOT All	All at home
Under \$50,000	36.27	21.08	10.65
\$50,000 - \$74,999	49.07	28.52	14.41
\$75,000 - \$99,999	35.2	20.46	10.34
Above \$100,000	39.47	22.94	11.59

$$X^2 = \sum \frac{(O-E)^2}{E} = \frac{(38-36.27)^2}{36.27} + \frac{(54-49.07)^2}{49.07} + \frac{(35-35.2)^2}{35.2} + \frac{(33-39.47)^2}{39.47} + \frac{(16-21.08)^2}{21.08} + \frac{(26-28.52)^2}{28.52} + \frac{(22-20.46)^2}{20.46} + \frac{(29-22.94)^2}{22.94}$$

$$+ \frac{(14-10.65)^2}{10.65} + \frac{(12-14.41)^2}{14.41} + \frac{(9-10.34)^2}{10.34} + \frac{(12-11.59)^2}{11.59}$$

$= 6.45$

df $= (r-1)(c-1)$
$= 3(2)$
$= 6$

$X^2_{0.05,6} = 12.592$

$\because$ $6.45 < 12.592$
$\therefore$ Non-reject Ho

b) $X^2 = 6.45$ df $= 6$
$X^2_{0.05,6} = 5.348$
$X^2_{0.25,6} = 7.841$
$\because$ $5.348 < 6.45 < 7.841$
$\therefore$ $0.25 <$ p-value < 0.5
$\therefore$ p-value > 0.05
$\therefore$ Non-reject Ho.